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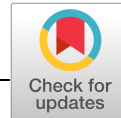
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ARTICLE

Understanding mAb aggregation during low pH viral inactivation and subsequent neutralization

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Abstract

Monoclonal antibodies (mAbs) and related recombinant proteins continue to gain importance in the treatment of a great variety of diseases. Despite significant advances, their manufacturing can still present challenges owing to their molecular complexity and stringent regulations with respect to product purity, stability, safety, and so forth. In this context, protein aggregates are of particular concern due to their immunogenic potential. During manufacturing, mAbs routinely undergo acidic treatment to inactivate viral contamination, which can lead to their aggregation and thereby to product loss. To better understand the underlying mechanism so as to propose strategies to mitigate the issue, we systematically investigated the denaturation and aggregation of two mAbs at low pH as well as after neutralization. We observed that at low pH and low ionic strength, mAb surface hydrophobicity increased whereas molecular size remained constant. After neutralization of acidic mAb solutions, the fraction of monomeric mAb started to decrease accompanied by an increase on average mAb size. This indicates that electrostatic repulsion prevents denatured mAb molecules from aggregation under acidic pH and low ionic strength, whereas neutralization reduces this repulsion and coagulation initiates. Limiting denaturation at low pH by D-sorbitol addition or temperature reduction effectively improved monomer recovery after neutralization. Our findings might be used to develop innovative viral inactivation procedures during mAb manufacturing that result in higher product yields.

KEYWORDS

ANS fluorescence, downstream processing, monoclonal antibodies, protein aggregation, protein unfolding, viral inactivation

1 | INTRODUCTION

Over the past decades, monoclonal antibodies (mAbs) and derived recombinant proteins have become an important class of active pharmaceutical ingredients and are used for the treatment of a substantial variety of diseases in humans (Elgundi, Reslan, Cruz,

Sifniotis, & Kayser, 2017; Grilo & Mantalaris, 2019; Kaplon & Reichert, 2018). At the same time, the biopharmaceutical industry has matured considerably since its inception in the 1980s, for example through the advent of generic drugs (so-called biosimilars; Udpa & Million, 2016). This evolution has led to increased cost pressure on manufacturing and fueled the desire for higher productivity (Gottschalk, Brorson, &

Shukla, 2012; Karst, Steinebach, & Morbidelli, 2018; Somasundaram, Pleitt, Shave, Baker, & Lua, 2018; Vogg, Wolf, & Morbidelli, 2018; Walther et al., 2015).

Therapeutic mAbs are commonly expressed in mammalian cells (Shukla, Hubbard, Tressel, Guhan, & Low, 2007; Vogg, Wolf et al., 2018), which intrinsically poses the risk of viral contamination. Chemical inactivation by temporary exposure to low pH is known to be a robust procedure for dealing with enveloped viruses (Brorson et al., 2003; Miesegaes, Lute, & Brorson, 2010) and is therefore an integral part of mAb manufacturing. Typically, protein A affinity chromatography is the first purification step for the clarified cell culture harvest (Somasundaram et al., 2018; Vogg, Müller-Späth, & Morbidelli, 2018). Elution of the mAb from the column is achieved by reducing mobile phase pH to a range from 3 to 4 (Hober, Nord, & Linhult, 2007; Pabst, Thai, & Hunter, 2018). Therefore, viral inactivation is conveniently performed immediately after protein A capture (Gillespie et al., 2019; Shukla et al., 2007). A typical viral inactivation protocol consists of reduction of the pH below 3.6 and incubation for at least 30 min with temperature $\geq 15^{\circ}\text{C}$ (ASTM International, 2012; Brorson et al., 2003). At the end of the viral inactivation step, the product stream has to be neutralized before it can be purified further (Carta & Jungbauer, 2010; Pfister, Nicoud, & Morbidelli, 2018).

Like virtually all proteins, mAbs are only marginally stable in their native, monomeric state in which they fulfill their biologic function (Roberts, 2014b, 2014a). Depending on solution conditions, mAbs have different tendencies to form nonnative protein aggregates (Wang, Nema, & Teagarden, 2010). Presence of such aggregates in the final drug product is a major concern due to their immunogenic potential (Filipe, Hawe, Schellekens, & Jiskoot, 2010). Therefore, mAb aggregates that are formed during manufacturing have to be removed by appropriate subsequent purification steps and represent a loss of valuable product (Pfister et al., 2018; Vázquez-Rey & Lang, 2011).

Exposure to acidic pH is known to denature mAb molecules and can lead to their aggregation (Arosio, Barolo, Müller-Späth, Wu, & Morbidelli, 2011; Arosio, Rima, & Morbidelli, 2013; Hari, Lau, Razinkov, Chen, & Latypov, 2010). Further, literature contains reports about mAb aggregation after temporary exposure to low pH followed by neutralization (Ejima et al., 2006; Filipe, Kükler, Hawe, & Jiskoot, 2012; Imamura & Honda, 2016; Oyetayo & Kiefer, 2016; Skamris et al., 2016). However, systematic investigation of viral inactivation process variables such as pH, ionic strength and, incubation time, and so forth on mAb self-association during low pH treatment as well as after neutralization is still lacking to the best of our knowledge. Ideally, this information would enable development of viral inactivation procedures that reduce the degree of incurred mAb aggregation and improve overall yield of the production process.

We systematically investigated the aggregation behavior of two mAbs under conditions similar to those encountered in the viral inactivation step during manufacturing. To simultaneously control pH and ionic strength of the solutions, an appropriate thermodynamic model was employed to accurately describe protonation equilibria of the buffer species. We used dynamic light scattering (DLS) and

1-anilinonaphthalene-8-sulfonate (ANS) fluorescence to evaluate mAb aggregation and conformational rearrangements at low pH, respectively. Monomer content and hydrodynamic size after neutralization was monitored by size-exclusion chromatography (SEC) and DLS, respectively.

This paper discusses the effects of pH, ionic strength and incubation time on mAb conformation and self-association during low pH incubation. Further, it describes the aggregation behavior after neutralization as a function of these variables. In addition, potential process strategies for improving monomer recovery after viral inactivation are presented. Those involve reduction of mAb denaturation at low pH either by use of an additive or by decreasing temperature.

2 | MATERIALS AND METHODS

2.1 | Materials

mAb-1, a monovalent antibody ($M_w = 100$ kDa) with pI equal to 7.3, was expressed by a proprietary Chinese hamster ovary cell line (Merck KGaA, Corsier-sur-Vevey, Vaud, Switzerland) cultivated inside a perfusion bioreactor. The harvest stream was purified using protein A affinity chromatography and immediately neutralized upon elution using 1 M Tris-HCl pH 8.0 buffer. The neutralized capture eluate was sterile-filtered using a 0.22 μm PES membrane syringe filter unit (TPP, Trasadingen, Zurich, Switzerland) and stored in the fridge at 4°C before use. mAb-2, an IgG-4 with pI equal to 7.9, was provided by Bristol-Myers Squibb (Devens, MA) as neutralized protein A capture eluate and stored frozen at -20°C before use.

Sodium phosphate monobasic, L-arginine, sodium azide, trisodium citrate, sodium chloride, sodium hydroxide, and D-sorbitol were purchased from Sigma-Aldrich. Citric acid was obtained from Fluka Analytical. ANS was bought from Acros Organics.

2.2 | Sample preparation and neutralization procedure

mAb stock solutions in ultrapure water at approximately 50 g/L protein concentration were prepared by ultrafiltration using centrifugal filter units followed by dialysis. First, an appropriate volume of neutralized eluate was reduced to 1 ml using an Amicon Ultra™ device (Merck Millipore, Burlington, MA) at 3000 rcf and 4°C . The concentrated eluate was transferred into a Slide-A-Lyzer™ dialysis cassette (Thermo Fisher Scientific, Waltham, MA) and dialyzed extensively against ultrapure water. The volume ratio of mAb solution to water was approx. 1:250. The water in the reservoir was exchanged four times over a period of 24 hr operating at 4°C under gentle agitation. The solution collected at the end of the dialysis was filtered through a 0.22- μm Millex-GV™ syringe filter (Merck Millipore) and stored in the fridge at 4°C before further use. mAb concentration of the stock solutions was determined by

injection into a Superdex 200 size-exclusion chromatography column coupled to UV absorbance detection at 280 nm (details provided below). In that way it was also confirmed that the mAb stock solutions always contained $\leq 3\%$ aggregates (on mass basis).

A total of 200 mM citric acid, 200 mM trisodium citrate, 1 M sodium chloride, 100 mM sodium hydroxide, and 2 M D-sorbitol stock solutions were prepared by dissolution of an appropriate mass of compound in ultrapure water and adjustment of the final solution volume using a volumetric flask. All stock solutions were filtered through 0.45 μm Durapore™ membrane filters (Merck Millipore) and stored in the fridge before use.

Samples measuring 500 μl , at desired mAb concentration, buffer molarity, pH, ionic strength, and D-sorbitol concentration, were prepared using 1.5 ml polypropylene reaction tubes (Sarstedt, Nümbrecht, Germany) by mixing appropriate volumes of mAb, citric acid, trisodium citrate, sodium chloride, and D-sorbitol stock solutions with the necessary volume of ultrapure water. Neutralization of samples to desired pH and ionic strength were performed by addition of 500 μl of neutralizing solution containing the required amount of sodium hydroxide and sodium chloride. Details of the calculation of the required stock solution volumes are presented in Section 2.6. Samples were mixed by pipetting up and down. Sample pH was always within ± 0.1 units of target value as independently verified by a Jenco 6230 N pH meter (Jenco Instruments, San Diego, CA). Experiments were performed at room temperature unless specifically stated otherwise. When temperature control was exercised, a Thermomixer® R 5355 block heating and cooling device (Eppendorf, Hamburg, Germany) and the G1330B autosampler thermostat (Agilent Technologies, Santa Clara, CA) of the chromatography system were used during low pH incubation and after neutralization, respectively.

2.3 | Dynamic light scattering measurements

Dynamic light scattering (DLS) measurements were performed on a Malvern Zetasizer Nano ZS instrument (Malvern Instruments, Malvern, Worcestershire, UK) equipped with a 633 nm He-Ne laser and detection at an angle of $\theta = 173^\circ$. A ZEN2112 quartz microcuvette (Malvern Instruments) was used for all measurements. For monitoring the aggregation kinetics, one measurement was performed every 5 min. A measurement consisted of 12 runs for evaluating the scattered intensity autocorrelation function, 10 s duration each. The laser attenuator was fixed at a value of 10 and the focal position of the laser inside the cuvette was fixed at 4.2 mm measuring from the outside of its front window.

2.4 | Size-exclusion chromatography

Chromatographic measurements were performed on an Agilent 1100 HPLC system (Agilent Technologies) consisting of degasser, quaternary pump, autosampler, and variable wavelength UV/Vis detector. A Superdex 200 10/300 GL size-exclusion column (GE Healthcare,

Chicago, IL) was used to conduct the chromatographic separations and all injections contained 50 μg of protein. The aqueous mobile phase contained 100 mM sodium phosphate pH 7.0 with 200 mM L-arginine and 1 g/L sodium azide and was applied at a flow rate of 0.5 ml/min. Chromatograms were recorded by UV-absorption at 280 nm wavelength.

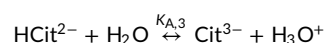
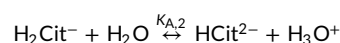
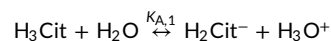
2.5 | ANS fluorescence measurements

ANS fluorescence in mAb solutions was measured on an EnSpire 2300 Multimode plate reader (Perkin Elmer, Waltham, MA) and 96-well nonbinding microplates with clear film bottom (Greiner Bio-One, Kremsmünster, AT). Fluorescence emission was recorded at 490 nm after excitation at 403 nm. Intensity was measured on a 3×3 grid within each well and averaged values were considered. ANS was dissolved in DMSO to obtain a stock solution at 2.5 mM concentration. Fluorescence measurements were conducted at 0.25 g/L mAb concentration and 10-fold molar excess of ANS (relative to the mAb). Samples were prepared as described above. The ANS stock solution was diluted 10-fold with ultrapure water and an appropriate volume was added to obtain the desired concentration.

2.6 | Thermodynamic calculations to control pH and ionic strength

Electrostatic interactions between protein molecules are controlled both by pH and ionic strength (Berg, 2009; Leckband & Israelachvili, 2001; Olsen, Andersen, Randolph, Carpenter, & Westh, 2009). To obtain meaningful results, both parameters should be controlled carefully during experiments. Matters are complicated due to the interconnection of the two quantities, which requires more detailed analysis of the relevant protonation equilibria as detailed here.

Throughout this study, citric acid was used as buffering agent to control solution pH. Citric acid (H_3Cit) can deprotonate three times as follows:



The thermodynamic equilibrium constant K_j for the j -th deprotonation step is given by

$$K_j = \frac{a_{\text{H}_3-j\text{Cit}^{j-}}^{\text{*,m}} a_{\text{H}_3\text{O}^+}^{\text{*,m}}}{a_{\text{H}_3-(j-1)\text{Cit}^{(j-1)-}}^{\text{*,m}} a_{\text{H}_2\text{O}}^0}, \quad (1)$$

where a_i represents the activity of species i . The superscript (*,m) indicates that the ideal dilute solution at $m = 1 \text{ mol/(kg solvent)}$, where m represents molality, is selected as reference state for the definition of activity (Luckas & Krissmann, 2001). On the other hand,

the superscript 0 denotes the pure liquid component i as reference state. The extended Debye–Hückel model was used to compute the activity coefficients $\gamma_i^{*,m}$ of the ionic species. It represents an empirical modification of the well-known Debye–Hückel limiting law and is valid up to approximately 100 mM of ionic strength. The single-ion activity coefficient is given by

$$\log_{10}(\gamma_i^{*,m}) = -A_m z_i^2 \frac{\sqrt{I_m}}{1 + d_i B_m \sqrt{I_m}}, \quad (2)$$

where z_i and d_i represent charge number and diameter of species i , respectively. Values for d_i are given in the supporting information (see Table S1). Parameters A_m and B_m take the value of $0.5108 \text{ kg}^{1/2} \text{ mol}^{-1/2}$ and $3.287 \times 10 \text{ m} \text{ kg}^{1/2} \text{ mol}^{-1/2}$ for water as solvent at 25°C, respectively (Luckas & Krissmann, 2001). Ionic strength is defined as $I_m = 1/2 \sum_k z_k^2 m_k$ with m_i being the molality of component i . Due to the limited effect of dissolved components on solution density, the difference between molality and molarity was neglected. Finally, the activity $a_i^{*,m}$ is computed as

$$a_i^{*,m} = \frac{\gamma_i^{*,m} m_i}{m^*} \cong \frac{\gamma_i^{*,m} c_i}{1 \text{ mol/L}}. \quad (3)$$

The activity of water was set to $a_{\text{H}_2\text{O}}^0 \cong 1$. These equations were used to compute the quantities of citric acid, trisodium citrate, sodium chloride, and sodium hydroxide required to obtain a solution of given pH and ionic strength.

3 | RESULTS AND DISCUSSION

3.1 | Behavior during low pH incubation

3.1.1 | Changes in molecular size

First, DLS was used to probe in-situ, whether the proteins aggregate under acidic conditions. Figure 1 shows the z-average hydrodynamic radius as a function of time for both investigated mAbs when incubated in 50 mM sodium citrate at pH 2.5, 3.0, and

3.5 with ionic strength fixed at 50 mM. As it can be expected based on their molecular weight (see Materials section), mAb-1 was slightly smaller than mAb-2 under each solution condition. For all cases, the average radius remained constant over the entire duration of low pH incubation. This result was not expected based on the various reports in literature on denaturation and aggregation of proteins under acidic conditions (Arosio et al., 2013; Hari et al., 2010). The absence of detectable aggregate formation can be explained by the low value of ionic strength deliberately selected to represent the situation encountered in downstream processing. Under acidic conditions, ionizable amino acid side chains, in particular of histidine, glutamic, and aspartic acid residues, are protonated to a significant extent. This implies a substantial positive protein surface charge. Electrostatic interactions between charged macromolecules immersed in a solution of mobile ions are to a great extent influenced by the ionic strength of the bulk solution. Quantitatively, the range of those interactions is proportional to the inverse of the Debye parameter κ , which is defined as (Berg, 2009)

$$\kappa = \sqrt{\frac{2000e^2 N_A^2 I}{\epsilon_0 \epsilon_r R T}}, \quad (4)$$

where e , N_A , I , ϵ_0 , ϵ_r , R , T represent elemental charge, Avogadro's constant, ionic strength, vacuum dielectric permittivity, solvent relative permittivity, universal gas constant, and system temperature, respectively. If electrostatic repulsion between mAb molecules is sufficiently strong and long-ranged, they have a very low probability of coming sufficiently close to allow for other interactions, such as those between hydrophobic residues, to prevail and lead to aggregate formation.

Next, the effect of ionic strength on mAb self-association during incubation under acidic conditions was investigated. For this, both mAbs were incubated in 50 mM sodium citrate at pH 3.0 with various ionic strengths and their average hydrodynamic radius was measured by DLS. The results shown in Figure 2 are qualitatively identical to those obtained at 50 mM ionic strength for different pH values, that is there was no statistically significant change in measured average mAb size over time. For mAb-1, hydrodynamic radius also

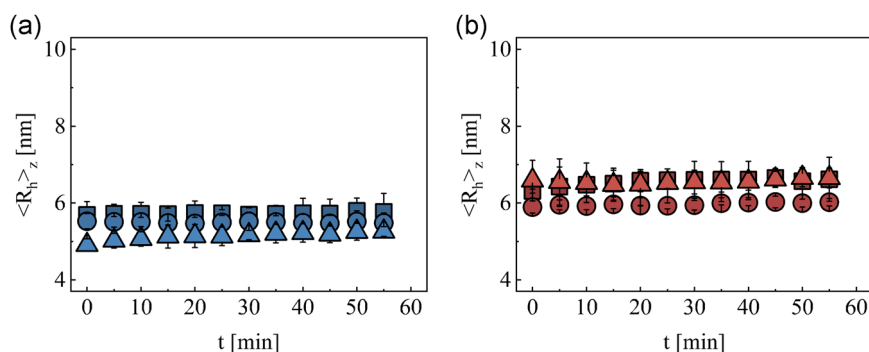


FIGURE 1 Average hydrodynamic radius measured by dynamic light scattering of mAb-1 (a) and mAb-2 (b) as a function of time during incubation at pH 2.5 (squares), 3.0 (circles), and 3.5 (triangles). 50 mM sodium citrate was used as buffer system and appropriate amounts of sodium chloride were added to fix the ionic strength at 50 mM. Symbols represent mean values of three independent repetitions and error bars correspond to 95% confidence intervals for the mean [Color figure can be viewed at wileyonlinelibrary.com]

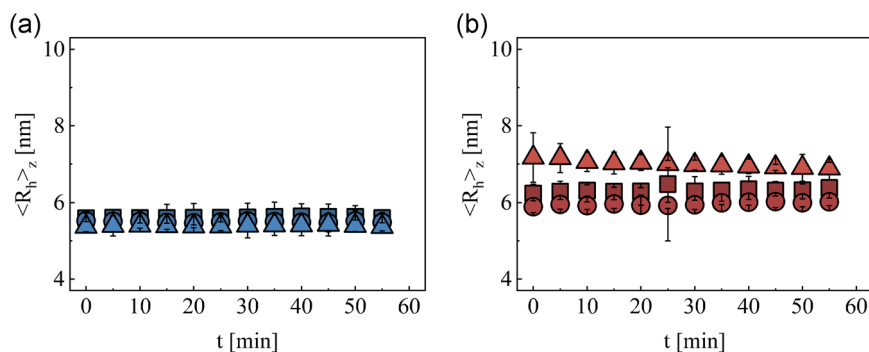


FIGURE 2 Average hydrodynamic radius measured by dynamic light scattering of mAb-1 (a) and mAb-2 (b) as a function of time during incubation in 50 mM sodium citrate pH 3.0 at 25 mM (squares), 50 mM (circles), and 100 mM (triangles) ionic strength. Symbols represent mean values of three independent measurements and error bars correspond to 95% confidence intervals for the mean [Color figure can be viewed at wileyonlinelibrary.com]

quantitatively did not depend on ionic strength in the range from 25 to 100 mM, while some variation in the level of R_h was observed for mAb-2, whose origin might be attributed to the measurement principle of dynamic light scattering. The detector of the instrument records in fact fluctuations in the intensity of the scattered light based on which the intensity autocorrelation function $G^{(2)}(\tau)$ is calculated (Lomakin, Teplow, & Benedek, 2005). This function is related to the normalized autocorrelation function for fluctuations in the electric field $g^{(1)}(\tau)$ through

$$G^{(2)}(\tau) = I_0^2(1 + \gamma(g^{(1)}(\tau))^2) \quad (5)$$

In this equation, I_0 represents the time-average of the scattering intensity and γ is the efficiency factor of the detector, which is related to its area. When fluctuations in the electric field are caused by Brownian motion of the scattering particles, the normalized electric field autocorrelation function shows an exponential dependence on the delay time τ given by

$$g^{(1)}(\tau) = \exp(-Dq^2\tau), \quad (6)$$

with q being the magnitude of the scattering wave vector and D the diffusion coefficient of the scatterers. In general, the diffusion coefficient shows a dependence both on q and τ , that is $D = D(q, \tau)$ (Doster & Longeville, 2007). However, for the length and time scales commonly probed by dynamic light scattering (i.e., low- q regime) it is simply equal to the collective diffusion coefficient D_c , which relates diffusive flux to gradients in chemical potential. Formally, it is given by

$$D_c = D_0 \frac{H(q \rightarrow 0)}{S(q \rightarrow 0)}, \quad (7)$$

where D_0 denotes the diffusivity of an individual scatterer at infinite dilution, which is related to its hydrodynamic radius R_h through the Stokes-Einstein relation (Berne & Pecora, 2000)

$$D_0 = \frac{k_B T}{f_0} = \frac{k_B T}{6\pi\eta R_h}. \quad (8)$$

In this expression, f_0 denotes the friction factor at infinite dilution and η represents the dynamic viscosity of the pure solvent. The hydrodynamic factor $H(q)$ captures “indirect,” hydrodynamic interactions between scatterers mediated through the solvent, whereas “direct” thermodynamic interactions are quantified by the static structure factor $S(q)$ (Blanco, Perevozchikova, Martorana, Manno, & Roberts, 2014). Both quantities depend on the pair interaction potential between two macromolecules, which in turn depends on the composition of the bulk solvent (Fine, Lomakin, Ogun, & Benedek, 1996). The ratio of those two quantities determines to which extent the measured diffusion coefficient D_c deviates from its value D_0 at infinite dilution. In simplified analysis of DLS data, as typically implemented in commercial instruments, it is assumed that measurement conditions always correspond to the dilute limit, that is D equals D_0 and Equation (8) is used to compute (average) hydrodynamic radii. In reality, interactions between scatterers can contribute to their observed diffusivity. Further, those interactions can change as a function of solution conditions. This would explain the different levels of R_h observed for mAb-2 at different ionic strengths.

Further, in the Supporting Information (see Figure S1) it is shown that the result in terms of ionic strength dependence of the hydrodynamic radius is qualitatively the same at pH 3.5 for both mAbs. In summary, it can be concluded that under conditions of low ionic strength, that is up to 100 mM, both mAbs do not aggregate at low pH.

3.1.2 | Changes in mAb surface hydrophobicity

Measurements of ANS fluorescence in mAb solutions were used to assess the extent to which the two mAbs undergo conformational rearrangements when incubated under acidic conditions, since the dye ANS can act as reporter for changes in protein surface hydrophobicity (Gasymov & Glasgow, 2007; Hawe, Sutter, & Jiskoot, 2008; Stryer, 1965). Experiments were performed with 50 mM sodium citrate as buffer system and at 50 mM ionic strength. The results are shown in Figure 3.

Fluorescence intensity stayed low and constant for both mAbs at pH 5.0, which is expected since the proteins are predominantly present in their native conformation under this condition and hydrophobic amino acid side chains are protected from solvent exposure inside the three-dimensional protein structure. For mAb-1 (Figure 3a), the fluorescence intensity gradually increases over the course of 60 min at pH 3.5 indicating that its surface hydrophobicity increases probably due to the progression of conformational rearrangements triggered by intramolecular charge–charge repulsion upon protonation of amino acid side chains. At pH 3.0 and 2.5, fluorescence is practically constant and slightly stronger than the value measured for pH 3.5 after 60 min. This indicates that conformational dynamics in response to exposure to those conditions very rapidly reach a steady state with significantly higher surface hydrophobicity compared to the reference situation at pH 5.0 and a slightly higher value compared to pH 3.5.

For mAb-2 (Figure 3b), fluorescence moderately increased over time at pH 3.5 relative to the control at pH 5.0 suggesting that protein surface hydrophobicity grew mildly due to conformational rearrangements of the mAb molecules. At pH 3.0, fluorescence intensity was constant over time at a higher level compared to the values measured at pH 3.5 throughout the entire duration of the experiment. Again, this indicates that conformational rearrangements promoted by the exposure to an acidic environment were virtually instantaneous, that is reached a steady state between preparation of the sample and first fluorescence measurement. Interestingly, increasing fluorescence over time was found again at pH 2.5, which started from a value similar to the level observed at pH 3.0 and reached more than twice this value after 60 min. Apparently, conformational dynamics of mAb-2 in response to exposure to pH 2.5 shows two phases. Immediately after sample preparation rapid rearrangements lead to a surface hydrophobicity similar to that observed at pH 3.0. Afterward, additional but slower conformational changes lead to further increase in solvent exposure of hydrophobic side chains.

Comparing the results for the two mAbs, lower ANS fluorescence values were recorded for mAb-2 at pH 3.5 and 3.0 compared to mAb-1, whereas the increase of fluorescence intensity over time

observed for mAb-2 at pH 2.5 led to higher values at later time points. The comparison indicates that the surface of mAb-2 is less hydrophobic relative to that of mAb-1 under less acidic pH. For both mAbs, ANS fluorescence showed a gradual increase over time at pH 3.5, which indicates that conformational changes caused by incubation at that pH value occur over time scales similar to typical duration of viral inactivation in mAb manufacturing. However, at lower pH, conformational dynamics can rapidly reach a new steady state.

3.2 | Behavior after neutralization

The results above suggest that under acidic conditions and low ionic strength the two mAbs (partially) denature and expose hydrophobic residues to the solvent. At the same time, electrostatic repulsion between mAb molecules due to the high protein surface charge and limited screening of electrostatic interactions by mobile ions prevents aggregation. However, mAb solutions have to be neutralized before subsequent processing steps, which will inevitably reduce their surface charge. Further, neutralization also increases ionic strength, which will shorten the range of electrostatic repulsion. The combination of these two effects could allow hydrophobic residues that became exposed at the protein surface in conformational rearrangements at acidic pH to interact and thereby lead to the formation of nonnative mAb aggregates. To study this process, the mAbs were incubated for different amounts of time at 50 mM ionic strength and pH 2.5, 3.0, or 3.5 (50 mM sodium citrate was used as buffer system in all cases). Afterwards, solutions were neutralized by addition of base and salt to pH 5.0 and 100 mM ionic strength and analyzed by SEC and DLS to monitor the loss of monomeric mAb due to aggregation and the growth of the aggregates, respectively.

Immediately after neutralization, 50 μ g of mAb solution were injected into the column and the injection was repeated once per hour. For illustrative purposes, Figure 4 shows SEC chromatograms recorded after neutralization for samples that were previously incubated for 40 min at pH 3.0. The large peak eluting after

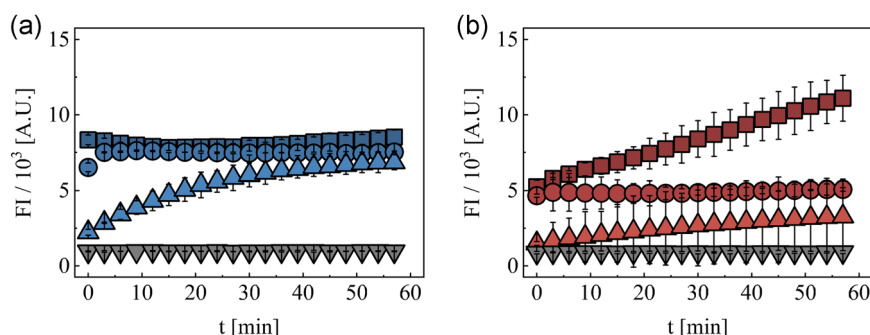


FIGURE 3 ANS fluorescence intensity of mAb-1 (a) and mAb-2 (b) solutions as a function of time during incubation at pH 2.5 (squares), 3.0 (circles), 3.5 (upwards triangles), and 5.0 (downward triangles). 50 mM sodium citrate with 50 mM ionic strength was used as buffer system for all experiments. Symbols represent mean values of two independent measurements and error bars correspond to 85% confidence intervals for the mean [Color figure can be viewed at wileyonlinelibrary.com]

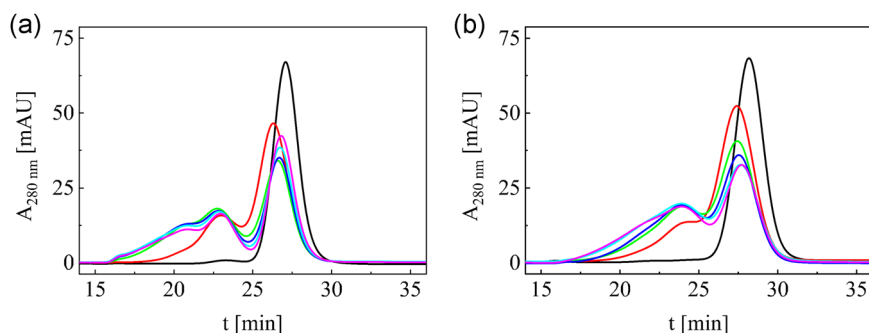


FIGURE 4 Size exclusion chromatography chromatograms of mAb-1 (a) and mAb-2 (b) solutions neutralized to pH 5.0. Before neutralization, monoclonal antibody (mAbs) were incubated at pH 3.0 for 40 min. Sodium citrate (50 mM) was used as buffer system with 50 and 100 mM ionic strength before and after neutralization, respectively. 50 μ g of protein were injected immediately after neutralization (red), 1 hr after neutralization (green), 2 hr after neutralization (blue), 4 hr after neutralization (cyan), and 8 hr after neutralization (magenta). The chromatogram for unstressed mAb (black) is shown as well [Color figure can be viewed at wileyonlinelibrary.com]

approximately 27 min corresponds to monomeric mAb, the peak eluting after approximately 23 min to dimeric mAb and larger oligomers are partially coeluting resulting in a convoluted peak from 15.5 to approximately 21 min. For mAb-1, the chromatogram obtained immediately after neutralization (red curve) showed mainly monomer and dimer with small quantities of larger oligomers. Subsequently, the peak area of the larger oligomers increased over time. Simultaneously, the peak area of the dimer started to decrease slightly after the second injection, that is for more than 2 hr after neutralization. At the same time, the peak area of the monomer increased during the later injections, the monomer peak became sharper and shifted to later elution times. Sharpening and shifting of the monomer peak could be explained by progressive refolding of denatured monomer molecules after neutralization, since partially unfolded conformations presumably have larger hydrodynamic size resulting in earlier elution. Overall, similar behavior was observed for mAb-2. In contrast to mAb-1, the monomer peak area decreased monotonically as a function of time as well as sharpening and shifting of the monomer peak was less pronounced.

Monomer relative to total peak area was used to determine the residual monomer content at each instant of time and the obtained values are shown in Figure 5 together with the average hydrodynamic

radius for both mAbs as a function of time after neutralization. It is seen that for mAb-1, the residual monomer fraction decreased within the first 2 hr after neutralization followed by a slight increase. At the same time, $\langle R_h \rangle_z$ initially rapidly increased but then decreased slightly at later times. For mAb-2, the fraction of monomeric protein decreased monotonically as a function of time after neutralization whereas the average hydrodynamic radius almost instantaneously reached a value more than twice as large as the one measured before neutralization (cf. Figure 1) and then remained constant.

As a next step, the effect of incubation time at low pH on the aggregation behavior after neutralization was investigated. Figure 6a–c show the residual monomer content for mAb-1 solutions as a function of time after neutralization for samples incubated for 20, 40, and 60 min at three different pH values: 2.5, 3.0, and 3.5, respectively. As described above, the monomer fraction dropped within the initial 2 hr after neutralization followed by a partial recovery for all considered conditions. At pH 2.5 and 3.0, incubation time under acidic conditions had no significant effect on monomer loss after neutralization. Further, monomer fraction as a function of time after neutralization followed the same time path for these two values of pH. Conversely, incubation time had an effect at pH 3.5 with longer duration giving lower values of monomer content at every time point after neutralization. These results are in very good agreement with the ANS fluorescence results

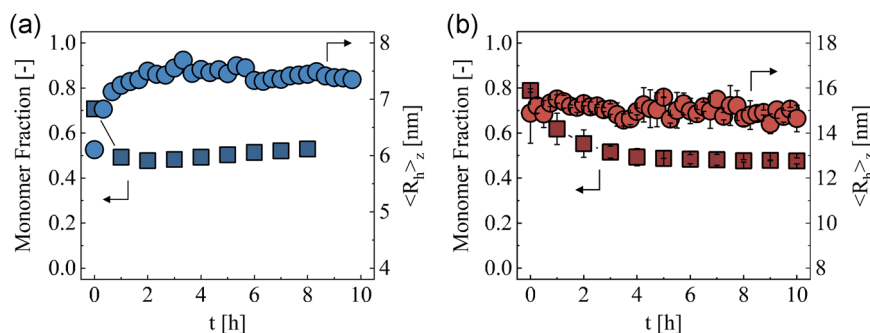


FIGURE 5 Fraction of residual monomeric protein (squares) and average hydrodynamic radius (circles) of mAb-1 (a) and mAb-2 (b) as a function of time after neutralization to pH 5.0 and 100 mM ionic strength. Before neutralization, monoclonal antibodies (mAbs) were incubated in 50 mM sodium citrate pH 3.0 for 40 min [Color figure can be viewed at wileyonlinelibrary.com]

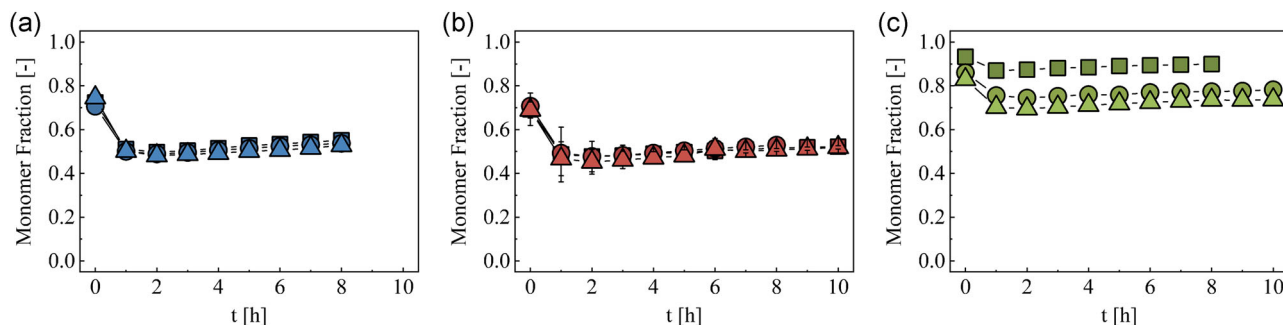


FIGURE 6 Fraction of monomeric mAb-1 as a function of time after neutralization to pH 5.0 and 100 mM ionic strength. Before neutralization the mAb was incubated for 20 (squares), 40 (circles), or 60 min at pH 2.5 (a), 3.0 (b), or 3.5 (c) and 50 mM ionic strength. Symbols represent mean values of two independent measurements and error bars correspond to 90% confidence intervals for the mean [Color figure can be viewed at wileyonlinelibrary.com]

presented in Figure 3. At the two lower pH values, fluorescence was almost identical and did not change as a function of time while at pH 3.5 fluorescence intensity gradually increased over the course of 60 min. Assuming that ANS fluorescence is proportional to the degree of denaturation, the extent of aggregation after neutralization corresponds to the degree of denaturation incurred at low pH.

Figure 7a–c show the kinetics of monomer loss after neutralization of mAb-2 for samples incubated for 20, 40, and 60 min at three different pH values: 2.5, 3.0, and 3.5, respectively. At pH 2.5 and pH 3.0, a monotonic decrease of the monomer fraction as a function of time after neutralization was observed in agreement with the results reported in Figure 5 whereas at pH 3.5 almost no aggregation was detected. Interestingly, the evolution of monomer content after neutralization was not affected by the incubation time at either pH value. Further, negligible differences between pH 2.5 and pH 3.0 in terms of aggregate formation were found. This observation does not fully agree with the ANS fluorescence results (cf. Figure 3), where gradually increasing fluorescence intensity was measured at pH 2.5 which was also stronger compared to that recorded at pH 3.0. It might be that the additional hydrophobic surface sites that become available at pH 2.5 compared to pH 3.0 are not involved in the formation of nonnative contacts between mAb molecules either for steric reasons or because the associated conformational changes are

rapidly reversible by neutralization. Although it cannot be expected that protein surface hydrophobicity as measured by ANS fluorescence and aggregation after neutralization are always perfectly correlated, it is worth noticing that the low ANS fluorescence in mAb-2 solutions at pH 3.5 is nicely coherent with the negligible amount of aggregate formation after neutralization.

The hypothesis that neutralization can lead to aggregation of mAb molecules that were partially denatured during previous incubation at low pH is strongly supported by the results presented in this section. In addition, it has to be mentioned that for conditions where substantial mAb aggregation was detected, the monomer fraction almost instantaneously dropped after neutralization, that is already the first SE chromatogram recorded immediately after neutralization (see Figure 4) showed significant quantities of oligomers. This leads to the conclusion that once electrostatic repulsion is abruptly reduced, coagulation proceeds rapidly, almost instantaneously, due to the strongly attractive hydrophobic interactions between surfaces of partially denatured mAb molecules (Imamura & Honda, 2016; Imamura, Sasaki, & Honda, 2017).

Presumably, there is a kinetic competition between refolding of denatured molecules and their self-association after neutralization. This conclusion, which refers to conditions typically encountered during viral inactivation in mAb manufacturing, is pictorially

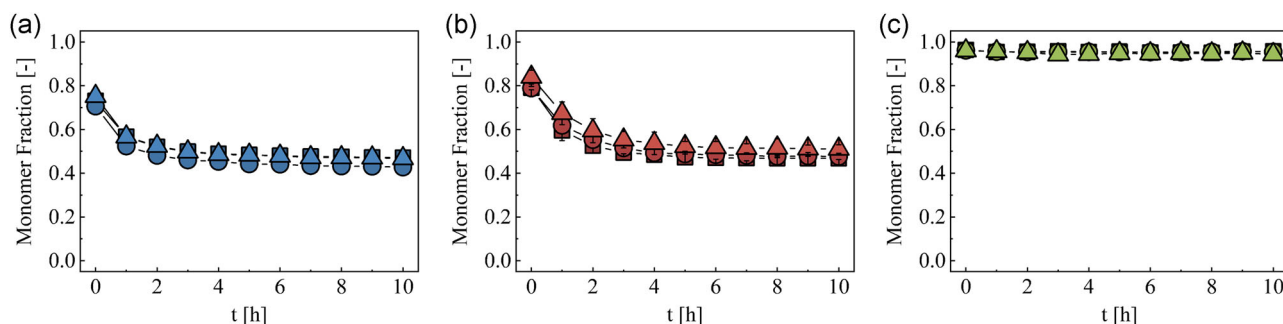


FIGURE 7 Fraction of monomeric mAb-2 as a function of time after neutralization to pH 5.0 and 100 mM ionic strength. Before neutralization the mAb was incubated for 20 (squares), 40 (circles), or 60 min at pH 2.5 (a), 3.0 (b), or 3.5 (c) and 50 mM ionic strength. Symbols represent mean values of two independent measurements and error bars correspond to 90% confidence intervals for the mean [Color figure can be viewed at wileyonlinelibrary.com]

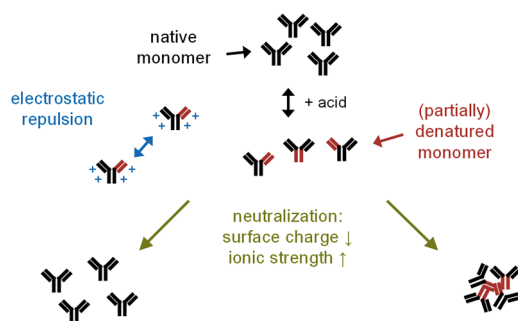
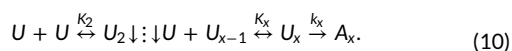


FIGURE 8 Graphical representation of the process of monoclonal antibody (mAb) denaturation and aggregation during viral inactivation. Exposure to an acidic environment leads to partial denaturation of mAb molecules. When ionic strength is sufficiently low during the low pH treatment, electrostatic repulsion prevents self-association of the partially denatured mAb. Neutralization of the solutions by addition of base lowers mAb surface charge and increases ionic strength, which leads to coagulation of denatured mAb molecules [Color figure can be viewed at wileyonlinelibrary.com]

summarized in Figure 8. More specifically, aggregation probably proceeds as follows: Upon exposure to low pH, the conformational equilibrium between the native state N and a (partially) unfolded state U shifts according to

$$N \xrightleftharpoons[k_f']{k_u'} U. \quad (9)$$

Here, k_u' and k_f' is the rate constant for the unfolding and folding reaction, respectively, and the prime indicates acidic conditions. As demonstrated, no oligomerization takes place under acidic conditions due to electrostatic repulsion. Neutralization then reduces this repulsion and enables the formation of oligomers as (Andrews & Roberts, 2007; Arosio, Rima, Lattuada, & Morbidelli, 2012; Li & Roberts, 2009)



It is assumed that the self-association of unfolded protein monomers is reversible up to a certain oligomer size x , followed by a structural rearrangement that renders the assembly irreversible (Andrews & Roberts, 2007; Li & Roberts, 2009). Irreversible aggregates then may grow further through addition of unfolded monomers or coagulation amongst itself. At the same time, (partially) unfolded molecules may convert back to the native state after neutralization according to

$$N \xrightleftharpoons[k_f]{k_u} U, \quad (11)$$

since k_u and k_f will take again their original values under neutral conditions with $k_f > k_f'$ and $k_f \gg k_u$. Indeed, mAb-1 showed partial reversibility of its aggregation over time (see Figures 5a and 6). In particular, the concentration of dimers and trimers appeared to

decrease after 2 hr (see Figure 4a). Further, based on these considerations, it can be expected that increasing the mAb concentration will increase the extent of aggregation, which was confirmed for mAb-1 (see Figure S3). On the other hand, mAb-2 did not show such reversibility, which proves that the details of the aggregation mechanism are highly mAb-dependent (Skamris et al., 2016). Collection of additional kinetic data would allow for more complete statements about the aggregation mechanism of the two molecules.

3.3 | Potential strategies for reducing MAb aggregation during viral inactivation

In this section we discuss how the improved understanding of mAb aggregation caused by temporary acidic treatment developed in this work can be used to develop strategies to mitigate aggregation during the viral inactivation step in mAb manufacturing. Three such strategies were tested and the results are presented below.

3.3.1 | Variation of neutralization velocity

As mentioned above, neutralization of acidic mAb solutions triggers two processes that compete with each other: refolding of partially denatured protein molecules and their self-association. So far, neutralization was performed by instantaneous addition of the entire volume of neutralization solution required to reach target pH and ionic strength values. Presumably, if refolding were given more time, while electrostatic repulsion between mAb molecules is still sufficiently strong, a lower proportion of the (partially) denatured mAb molecules would be incorporated into aggregates. Accordingly, we investigated, if slower addition of the neutralizing solution affects the extent of monomer loss due to aggregation.

To test this hypothesis, both mAbs were incubated for 40 min in 50 mM sodium citrate pH 3.0 at 50 mM ionic strength. Subsequent neutralization to pH 5.0 and 100 mM ionic strength was performed over predetermined time intervals through continuous addition of the neutralizing solution with a syringe pump under gentle agitation. After completing the addition, solutions were analyzed by SEC over ten consecutive hours. Results are presented in Figure 9. It is seen that, for both mAbs, the duration of neutralization, in the range from 0 to 60 min, had minimal impact on monomer depletion over time. Interestingly, after neutralizing the mAb solutions over the course of 10 hr, a constant level of monomer fraction was observed over time. Its value was equal to the final residual monomer fraction obtained for the instantaneous neutralization protocol (cf. Figures 6 and 7).

These observations lead to the conclusion that for neither mAb a slower increase in pH and ionic strength has a beneficial effect with respect to the final extent of aggregate formation. This circumstance could be explained either by the fact that partial denaturation incurred at low pH is not reversible solely by increasing pH, which simply reduces intramolecular charge–charge repulsion (Martsev, Kravchuk, Vlasov, &

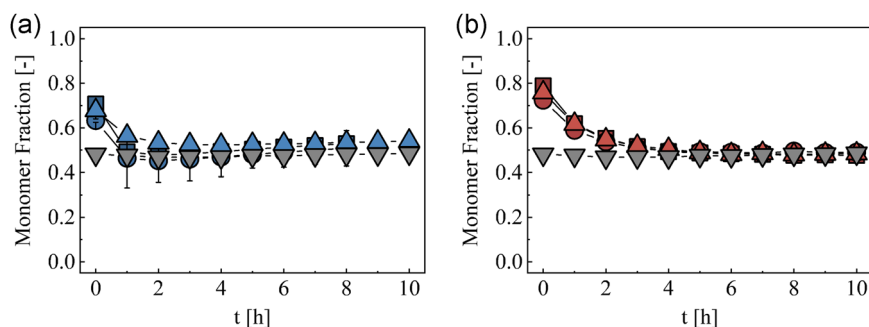


FIGURE 9 Fraction of residual monomeric mAb-1 (a) and mAb-2 (b) as a function of time after completed neutralization to pH 5.0 and 100 mM ionic strength. Monoclonal antibodies (mAb) were incubated for 40 min in 50 mM sodium citrate pH 3.0 at 50 mM ionic strength before neutralization. The neutralizing solution was added immediately (squares), over 30 min (circles), over 60 min (upward triangles), or over 10 hr (downward triangles) [Color figure can be viewed at wileyonlinelibrary.com]

Lyakhnovich, 1995; Rowe et al., 2018), or refolding is invariably too slow to prevent aggregation. As a consequence, the fraction of protein that denatured at low pH will inevitably aggregate, once electrostatic repulsion between those molecules is sufficiently reduced.

3.3.2 | Limiting extent of denaturation at low pH: Additives

Another potential strategy to mitigate the loss of monomeric mAb by low pH treatment for viral inactivation might be to limit the extent of mAb denaturation that occurs under acidic conditions. This could be achieved through the addition of polyol sugars, which are known to be preferentially excluded from the protein surface in solution (Timasheff, 1993, 1998). As a consequence, they are capable of shifting the conformational equilibrium towards the native state, since the chemical potential of the denatured protein increases to a larger extent compared to that of the native state in presence of polyol sugars. D-Sorbitol has already been shown to slow down heat-induced aggregation of two monoclonal antibodies (Nicoud, Cohrs, Arosio, Norrant, & Morbidelli, 2015; Nicoud et al., 2014) and was selected as model compound.

Specifically, the effect of D-sorbitol on mAb-1 surface hydrophobicity and aggregation behavior was studied at pH 3.5. ANS fluorescence in mAb-1 solutions was measured as a function of time in presence of 0, 250, and 500 mM D-sorbitol. Further, residual monomer fraction was monitored by SEC after standard neutralization to pH 5.0 after 60 min incubation at pH 3.5. Results are presented in Figure 10. Fluorescence intensity showed qualitatively the same gradual increase over time for all three D-sorbitol concentrations (Figure 10a). Quantitatively, fluorescence decreased as a function of D-sorbitol concentration for any given point in time. This suggests that mAb surface hydrophobicity and therefore its degree of unfolding are reduced by the presence of D-sorbitol in a concentration-dependent manner. However, it must be mentioned that the substantial variability of the ANS assay rendered the differences in fluorescence intensity insignificant from a statistical point of view. Additionally, Figure 10b shows that monomer loss after neutralization also decreased as a function of added amount of D-sorbitol during low pH incubation. These observations agree well both with the proposed mechanism of mAb aggregation as well as with the impact of D-sorbitol on mAb conformational equilibrium and could represent a method to improve product yield within the context of viral inactivation.

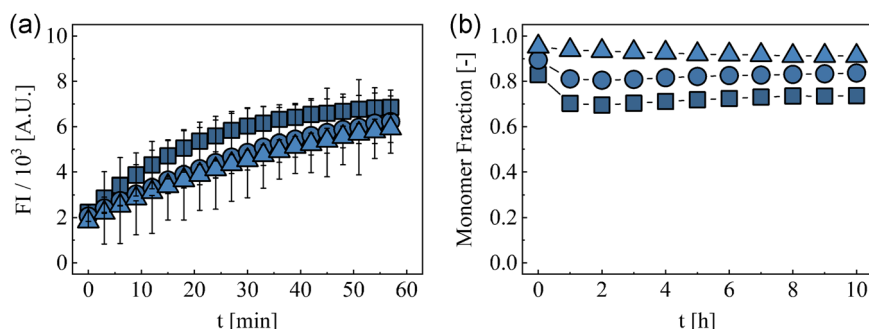


FIGURE 10 (a) 1-Anilinonaphthalene-8-sulfonate fluorescence of mAb-1 solutions as a function of time at pH 3.5 and 50 mM ionic strength in the presence of 0 (squares), 250 (circles), and 500 mM (triangles) D-sorbitol. (b) Time evolution of the residual monomer fraction of mAb-1 after neutralization to pH 5.0. Before neutralization, the monoclonal antibody (mAb) was incubated for 60 min at pH 3.5 in presence of increasing quantities of D-sorbitol. Symbols correspond to the same concentrations as in (a) [Color figure can be viewed at wileyonlinelibrary.com]

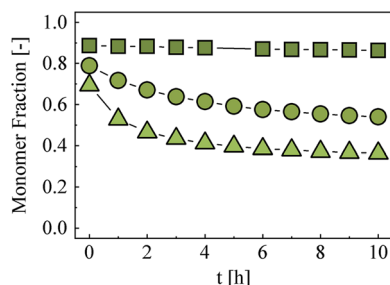


FIGURE 11 Time course of the monomer fraction of mAb-2 solutions after neutralization to pH 5.0. Before neutralization, mAb-2 was incubated at pH 3.25 for 60 min. During low pH hold and after neutralization the temperature was kept at 5°C (squares), 20°C (circles), or 35°C (upward triangles) [Color figure can be viewed at wileyonlinelibrary.com]

It has to be mentioned that under harsher conditions (i.e., pH 3.0), addition of D-sorbitol did not lead to differences in aggregation behavior after neutralization (see Supporting Information). However, such low pH values are not commonly employed in mAb manufacturing (ASTM International, 2012; Brorson et al., 2003; Miesegaes et al., 2010).

3.3.3 | Limiting extent of denaturation at low pH: Reducing temperature

Protein denaturation is known to be strongly affected by temperature as well (Becktel & Schellman, 1987; Edelhoch & Osborne, 1976). To test the influence of temperature on the mAb aggregation during viral inactivation we incubated mAb-2 at pH 3.25 for 60 min followed by the standard neutralization protocol to pH 5.0. We kept the temperature at a fixed value, both during low pH incubation and after neutralization. The results in terms of monomer fraction after neutralization are presented in Figure 11. Both at 20°C and 35°C, substantial monomer loss was observed over time after neutralization. However, lowering the temperature further to 5°C strongly improved the monomer recovery from 54% (at 20°C) to 86% at 10 hr after neutralization. Further, the monomer fraction after neutralization stayed constant over time at 5°C. Decreasing pH effectively lowers the denaturation (or melting) temperature of mAbs (Buchner et al., 1991; Privalov, 1982). The fraction of (partially) denatured monomers that are formed at low pH will depend on the difference between the denaturation temperature and the actual temperature. By lowering the temperature, one can reduce the extent of incurred denaturation during low pH incubation and, in line with the arguments presented above, improve the monomer yield after neutralization.

4 | CONCLUSIONS

Under conditions of low pH and low ionic strength, mAb molecules partially denature, but do not aggregate due to the sufficient

electrostatic repulsion between them. Subsequent neutralization raises pH and ionic strength, which reduces this repulsion and enables self-association of partially denatured monomers. Extrinsic fluorescence can be used to monitor extent and time evolution of denaturation at low pH and there is a partial correlation with the amount of aggregates observed after neutralization. Limiting the extent of mAb denaturation incurred under acidic conditions (e.g., through the use of suitable additives or reduction of temperature) can increase the residual monomer content obtained after neutralization.

Formation of mAb aggregates after transient exposure to acidic conditions has been reported in the literature (Ejima et al., 2006; Filipe et al., 2012; Imamura & Honda, 2016; Oyetayo & Kiefer, 2016; Skamris et al., 2016). To our knowledge, this is the first study that systematically controlled both pH and ionic strength during low pH incubation as well as after neutralization. Overall, our observations are in line with literature reports for other mAbs under comparable conditions (Bickel et al., 2016; Jin et al., 2019; Mazzer, Perraud, Halley, O'Hara, & Bracewell, 2015; Skamris et al., 2016). Therefore, we believe that our conclusions will hold for the majority of mAb molecules.

Further, the presented results could be useful for the development of viral inactivation procedures that reduce product loss due to aggregate formation. We found that the temperature at which the viral inactivation is performed has a strong influence on the fraction of monomeric mAb that is lost due to aggregation. Moreover, future work will be directed at identifying additional compounds that are more potent than D-sorbitol in reducing mAb denaturation at low pH, since the use of additives might be an alternative route to improved monomer recoveries.

At the same time, the influence of any process modification on the viral inactivation efficacy must be considered as well. Fundamentally, reduction of virus infectivity upon exposure to acidic conditions is, at least in part, caused by denaturation of virus envelope proteins (Brorson et al., 2003; Edwards, Mann, & Brown, 1983; Weed, Pritchard, Gonzalez, Aguilar, & Nicola, 2017). Thus, there will probably always be a trade-off in terms of operating conditions between inactivation efficacy and product stability. However, the inactivation kinetics of a widely used model virus were shown to be moderately sensitive to temperature (Brorson et al., 2003; Chinniah, Hinckley, & Connell-Crowley, 2016; Gillespie et al., 2019), which could render temperature reduction as a valid approach for improving monomer recovery. On the other hand, the effect of additives like D-sorbitol on the viral inactivation efficacy would have to be evaluated first. Further, general compatibility of any measure with the entire mAb manufacturing process has to be evaluated as well. Moreover, improvements in process yields must economically justify any modification to the production process.

A fundamentally different approach would be (re-)engineering of the product's primary structure to increase its tolerance towards acidic conditions (Ducancel & Muller, 2012; Saito, Namisaki, Hiraishi, Takahashi, & Iida, 2019; Skamris et al.,

2016). We have shown that a simple method such as ANS fluorescence could be used to identify candidate molecules that have a low propensity to denature under (mildly) acidic conditions and would thus aggregate to a low extent during subsequent neutralization.

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CONFLICT OF INTERESTS

The authors declare that there are no conflict of interests.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section.

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